

Polly — Pre-MVP3 Research

Executive briefing (5-page) · Polymarket trading-strategy & data program · primary-source-verified 2026-06-25

1 · Strategic conclusion

Detecting informed/insider activity is easy; capturing it is the hard part — and following a confirmed fill is a losing race. Once a sharp trade prints, its permanent price impact (Kyle 1985) is incorporated within seconds, the public WebSocket reveals trade direction only ~59% of the time (Dubach 2026, panel mean 0.615), MEV/HFT bots fill in milliseconds, and Polymarket charges a category-based taker fee on a $p(1-p)$ curve. The insider's fill is the signal, so a follower captures only the decayed residual minus fees and slippage.

MVP3 therefore abandons leaderboard-copying (the mvp1/mvp2 verdict: not a robust edge after latency, fees, and survivorship) and pivots to two things that compound: **(a) owning the full point-in-time information set** Polymarket exposes, and **(b) trading signals that do not self-erase when observed.**

The four defensible edges (shipped in robustness order)

- **Passive market-making** VERIFIED — Polymarket *subsidizes* makers: no maker fee, a 20% (crypto) / 25% (other) rebate of taker fees, plus quadratic near-mid liquidity rewards. The most structurally robust candidate; but passive LP is *underwriting*, not spread capture, so cap resolution exposure and pull quotes around event conclusion.
- **Mispricing / fair value** — de-vig and ensemble external probability anchors and bet the deviation. Strongest for sports (Shin de-vig of consensus odds) and long-horizon politics (compression-toward-50% correction). Internal arbitrage is real (~\$40M/yr extracted) but capacity-bound to ~15 shares.
- **Fade the dumb money** VERIFIED — a ~3% skilled minority drives price discovery and captures >30% of gains; the unskilled majority funds them. Unlike sharp-following, this signal is **large, persistent and not self-erasing**.
- **News-latency** — react to news faster than slow humans inside the ~0.64 passthrough window. The one genuine fast race; expect MEV/HFT competition and validate hard.

Insider detection is used defensively — as a toxicity filter for market-making and a fade confirmation — never as a copy signal.

Verification verdict: the strategic conclusion survived primary-source re-fetch unchanged. The only reversal (favorite-longshot bias is classic, not reversed) reinforces the fair-value edge rather than weakening it. All corrections below are to numbers and citations, not the thesis.

2 · Pillar A — Insider / informed-trading detection

It happens, it's real, and it's prosecutable: a DOJ-charged US soldier turned ~\$33k into ~\$410k on a classified-info bet; a Bubblemaps 9-wallet cluster hit a **98% win rate over 80+ bets (\$2.4M)** on US-military markets (accounts created days before the event); Columbia found ~25% of volume (~45% of sports) is wash-traded. The "Théo" whale (11 wallets, >\$85M) was a **research-driven false positive** — the central design warning.

The flagship screen — **Mitts & Ofir 2026, "From Iran to Taylor Swift"** — over >210,000 wallet-market pairs flags a **69.9% win rate / ~\$143M anomalous profit** (>60 SD above a permutation null) by combining bet-size anomaly + profitability + pre-event timing + directional concentration.

Engineer these on-chain signatures

- **Wallet freshness** (age <48h, low nonce, first trade >\$1k); **funding-source clustering** via the EVM deposit-address heuristic (Victor 2020), not Bitcoin co-spend; **concentration**; **last-minute one-sided taking**; **anomalous win-rate** ($p < 0.001$).
- Transfers from TradFi: event-study runoff/CAR (Meulbroek 1992 — insider-day move ≈40–50% of the public reaction), PIN, and VPIN as a toxicity gauge (its flash-crash predictive claim is contested).
- **Composite scoring is mandatory** to avoid Théo-style false positives. **CORRECTED** The earlier "GNN/GAT 95–100% Sybil accuracy" claim is **unsupported** and was dropped; rely on funding/gas-graph + behavioral heuristics.

3 · Pillar B — Mispricing / fair value

Polymarket is well-calibrated near resolution (Brier **0.0627** self-reported) but **compressed toward 50% at long horizons**: a 70¢ politics contract a month out implies a true ~75% **CORRECTED** (Page & Clemen, *Economic Journal* 2013; Le 2026). **CORRECTED** The favorite-longshot bias is the **classic** direction (favourites underpriced, longshots overpriced) — *not* reversed; this is the same compression effect, so tilt the standard way.

- **Internal arbitrage**: ~\$40M extracted in one year, but per-opportunity median **101 bps**, constrained to ~**14.8 shares** (76.9% of cases), median life **3.6s**. **DOWNGRADED** The "half-life<1min" and cross-venue "6% / 2–4% / ≥1h" figures were unsupported and removed.
- **External anchors**: markets beat polls ~74% long-horizon; 538-type models roughly tie the market; Shin's method is the best de-vig of sportsbook odds; superforecasters beat markets 15–30% on Brier.
- **Dynamics**: ~0.64 news passthrough with multi-minute drift; overreaction to surprises; ~50% open-to-close reversal in pre-game lines (Moskowitz 2021); volatility rises into resolution.

4 · Pillar C — Microstructure

- **OFI predicts short-term moves** (Cont-Kukanov-Stoikov, arXiv 1011.6402: avg R^2 **65%** for order-flow imbalance vs **32%** for trade imbalance) **CORRECTED** id & figures; edge decays in ~10s → a maker/queue-positioning signal, not taker arbitrage.
- **Stoikov micro-price** (mid + imbalance + spread; a martingale) is the best short-horizon fair value.
- **Critical caveat:** public-feed trade direction matches on-chain ground truth only ~59% (vs ~80% Lee-Ready on Nasdaq), and the effective half-spread flips sign on 50–67% of markets → **signed-flow signals must be built from on-chain data**.
- Depth ≈ uniform grid; **CORRECTED** spreads are measured in **bps** (~400 bps central, 1,300–1,800 bps below 0.10), not "2–5¢"; the "3.5× Kalshi" figure is unverified.
- **Frozen-liquidity hazard:** NBA post-game spreads exploded to **7,532 bps** and 81% of theoretical arbs were post-game/unexecutable — pull quotes around event conclusion.

5 · Pillar D — Data sources & collector architecture

- **On-chain (Polygon):** CTF ERC-1155 (stable); CTF Exchange `OrderFilled` is the trade tape — attribute per-fill on `maker`, never the relayer. **CORRECTED** **v2 `OrderFilled` is a NEW event** (distinct topic0 + separate `FeeCharged`) → **a dual decoder is mandatory**; index both exchange generations and both collateral tokens (USDC.e → pUSD). All addresses confirmed verified on Polygonscan.
- **CORRECTED** **UMA fast-path liveness = 2h** (the 48h figure is the DVM voting phase, only on escalation).
- **APIs:** Data API is the real-time backbone; CLOB `/book` + WebSocket for order books (REST L2 backfill stopped ~Feb 2026 → capture live); Goldsky/Dune for historical joins.
- **Off-chain anchors:** GDELT (free, 15-min), The Odds API (de-vig, 500 credits/mo free), ESPN JSON. **CORRECTED** **FiveThirtyEight shut down (Mar 2025) — dropped**; no free raw-poll replacement.
- **Storage:** bitemporal (valid-time + ingest-time) for leakage-free as-of queries; Parquet/DuckDB + ClickHouse; ~30-block confirmation buffer (Polygon max reorg ≈32 blocks).

6 · Pillar E — Forward-test methodology

- Forward > historical: fixing the universe at t0 kills survivorship; point-in-time decisions kill look-ahead. The ceiling: paper trading can't model own market impact ($\sqrt{Q}$ law), reflexivity, or true slippage.
- **Required sample scales as $1/SR^2$.** VERIFIED PSR, MinTRL, Deflated Sharpe, and Lo's Sharpe SE formulas byte-verified against reference implementations.
 - **Report in stability order:** win-rate first ($\approx 5pp$ at ~ 100 trades), Sharpe provisional (slow), max drawdown explicitly provisional (biased, grows with window).
 - **Peek without inflation:** CORRECTED monthly looks inflate Type-I error to ~ 0.14 at 5 looks / ~ 0.19 at 10; use O'Brien-Fleming alpha-spending or, better, **e-values / confidence sequences** (anytime-valid). Report a **Deflated Sharpe** (Harvey-Liu-Zhu hurdle $t > 3.0$).

Verification corrections at a glance

Claim	Was	Now (primary-verified)
Favorite-longshot bias	reversed on Polymarket	classic direction (Le 2026) — reinforces fair-value edge
Long-horizon example	70¢ a week out $\approx 83\%$	70¢ a month out $\approx \sim 75\%$; cite <i>Economic Journal</i> 2013
Arb per-opportunity	$\sim 100bps$, half-life $< 1min$	median 101 bps, 14.8 shares, 3.6s ; half-life claim removed
Cross-venue arb	6% listed, 2–4% $\geq 1h$	unsupported → speculation (no source studies it)
OFI	arXiv 1011.4087, 44/50	arXiv 1011.6402 , avg R^2 65% vs 32%
Spreads	2–5¢; 3.5× Kalshi	measured in bps ; 3.5× Kalshi unverified
v2 OrderFilled	same as v1	new topic0 + FeeCharged → dual decoder
UMA liveness	2h vs 48h unclear	2h fast path; 48h DVM-only
538 poll feed	continuity uncertain	dead (shut down Mar 2025) → dropped
Multiple-looks inflation	~ 0.19 at 5 looks	~ 0.14 at 5, ~ 0.19 at 10
GNN/GAT Sybil 95–100%	load-bearing	unsupported → dropped

7 · The MVP3 build (design handoff)

Architecture

Six bitemporal collector streams (L2 books, trade tape, on-chain flows, resolutions, off-chain anchors, market metadata) feed a raw immutable log → bitemporal store → a **point-in-time as_of(t)** API that is the only thing the decision layer may call. This two-layer firewall is the look-ahead defense mvp1/mvp2 lacked.

Strategy set (robustness order, each a pure function of the point-in-time feature store)

#	Strategy	Why it's robust / the catch
A	Passive market-making	Subsidized makers + biased flow; risk = underwriting → pull quotes near conclusion, avoid toxic flow
B	Mispricing / arbitrage	Structural but capacity-bound (~15 shares); fair-value deviation strongest for sports & long-horizon politics
C	Fade the dumb money	Large, persistent, non-self-erasing; pairs naturally with market-making
D	News-latency	Genuine fast race in the 0.64 window; thin, MEV/HFT competition
E	Insider/toxicity (defensive)	Toxicity filter + fade confirmation; composite score, never a copy signal

Forward test — the deliverable

Paper-trade in real time, simulate fills by walking the recorded L2 (not mid) with a latency-Δ, charging exact category fees. Evaluate at **1 / 2 / 6 months** with anytime-valid statistics. A strategy graduates to a real-capital sleeve only if, at 6 months: anytime-valid $p < 0.05$ (or e-value > 20) net of fees/slippage, win-rate CI excludes the null, Deflated Sharpe > 0 , and observed capacity $\geq$ a usable bankroll. Otherwise document and retire.

Build order

- 1 · Collector MVP (books + trades + resolutions + metadata, bitemporal, t0 universe fixed) — start recording immediately (the 6-month clock starts at capture).
- 2 · On-chain flow + funding-graph + wallet features (insider/toxicity score, EB skill).
- 3 · Fair-value engine (de-vig/ensemble) + on-chain microstructure features.
- 4 · Paper-trading harness + strategies A → E behind the point-in-time firewall.
- 5 · 1/2/6-month forward evaluation with anytime-valid stats; promote/retire.

Operational note: the always-on collector must run on persistent infrastructure (a small VPS + object storage) for the multi-month window; the repo ships the code, schema, and run-book.

Full detail and per-claim verification in `research/PRE_MVP3_RESEARCH.md`, `research/MVP3_DESIGN.md`, and `research/VERIFICATION_LOG.md`. Tags: **VERIFIED** primary-source byte-verified · **CORRECTED** number/citation fixed · **CORRECTED** claim reversed · **DOWNGRADED** to unverified/speculation. A minority of items remain cross-checked search snippets where publishers (SSRN, ResearchGate, Yale Insights) return origin anti-bot blocks.